

Create Service

1-Create service

>> ng g s services/countries

```
1 import { Injectable } from '@angular/core';
2 import { HttpClient } from '@angular/common/http'
3
4 @Injectable({
5   providedIn: 'root'
6 })
7 export class CountriesService {
8
9   constructor(private _httpClient:HttpClient) { }
10
11   public getCountries():any{
12     return this._httpClient.get("http://localhost:8080/api/countries");
13   }
14 }
15
```

- @Injectable: "This class is a service and can be injected into other classes."
- import {HttpClient} from '@angular/common/http' យក HttpClient មកប្រើប្រាស់សម្រាប់ Rest API (GET, POST, PUT, DELETE)
- constructor(private _httpClient:HttpClient) { } : Constructor + Dependency Injection. ពេលវាចាំឱ្យដំណើរការ វា បង្កើតមុនគេ (Constructor runs when Angular creates the service.)
- private _httpClient: This creates a private variable.
- public getCountries(): any {} = This creates a function កន្លែងដែលយើងដាក់ API

Flow:

```
CountriesService
  ↓
needs HttpClient
  ↓
Angular creates HttpClient
  ↓
Angular injects it
```



2- Create component

>> ng g c components/countries

```
import { Component, OnInit } from '@angular/core';
import { HttpResponse } from '@angular/common/http';
import { CountriesService } from 'src/app/services/countries.service';

@Component({
  selector: 'app-countries',
  templateUrl: './countries.component.html',
  styleUrls: ['./countries.component.css']
})
export class CountriesComponent implements OnInit {

  public data: any[] = [];

  constructor(private _service: CountriesService) { }

  ngOnInit(): void {
    this._service.getCountries().subscribe({
      next: (response: any[]) => {
        this.data = response;
      },
      error: (err: HttpResponse) => {
        console.error('Error loading countries:', err);
      }
    });
  }
}
```

- OnInit: Run code when component starts.
- import { HttpResponse } from '@angular/common/http' : Used for API error handling.
- constructor(private _service: CountriesService) { } : Inject the service
- public data: any[] = []; ประกาศ variable

3- HTML

```
<h1>Table List Country</h1>
<table>
  <thead>
    <tr>
      <th>Name</th>
      <th>Capital</th>
      <th>Region</th>
      <th>Calling Codes</th>
    </tr>
  </thead>
  <tbody>
    <tr *ngFor="let c of data">
      <td>{{c.name}}</td>
      <td>{{c.capital}}</td>
      <td>{{c.region}}</td>
      <td>{{c.callingCodes[0]}}</td>
    </tr>
  </tbody>
</table>
```

4- View it: app.module.ts

```
import { NgModule } from '@angular/core';
import { BrowserModule } from '@angular/platform-browser';

import { AppComponent } from './app.component';
import { CountriesComponent } from './components/countries/countries.component';
import { CountriesService } from './services/countries.service';
import { HttpClientModule } from '@angular/common/http';

@NgModule({
  declarations: [
    AppComponent,
    CountriesComponent
  ],
  imports: [
    BrowserModule,
    HttpClientModule
  ],
  providers: [CountriesService],
  bootstrap: [CountriesComponent]
})
export class AppModule { }
```

- providers: [CountriesService] យើងដាក់ Service ចូលក្នុង providers
- bootstrap: [CountriesComponent]: កន្លែងដែល កម្មវិធីចាប់ផ្តើម starter app ចឹងបើយើងដាក់ component អីវានឹងបង្ហាញអានឹង

5- Index.html

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <title>Countries</title>
  <base href="/">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="icon" type="image/x-icon" href="favicon.ico">
</head>
<body>
  <!-- <app-root></app-root> -->
  <app-countries></app-countries>
</body>
</html>
```

- <app-countries></app-countries> inside the body we put the component where we want to display